

Program 10

Step 1:- Launch Kubernetes Dashboard

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog9$ minikube dashboard
```

```
🤖 Verifying dashboard health ...
```

```
🚀 Launching proxy ...
```

```
🤖 Verifying proxy health ...
```

```
🌐 Opening http://127.0.0.1:40499/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard/proxy/
```

```
Gtk-Message: 09:58:22.956: Not loading module "atk-bridge": The functionality is provided by GTK
```

Step 2:- Create deployment.yaml and service.yaml

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog10$ cat deployment.yaml
```

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
```

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog10$ cat service.yaml
```

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort
```

Step 3:- Apply deployment and service.yaml

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog10$ kubectl apply -f deployment.yaml
kubectl apply -f service.yaml
deployment.apps/nginx-deployment created
service/nginx-service created
```

Step 4:- Monitor Using CLI

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog10$ kubectl get pods
kubectl get services
kubectl get deployments
```

NAME	READY	STATUS	RESTARTS	AGE
nginx-deployment-6f9664446b-4hgdq	1/1	Running	0	10s
nginx-deployment-6f9664446b-ptz9d	1/1	Running	0	10s

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	13h
nginx-service	NodePort	10.105.247.242	<none>	80:31878/TCP	10s

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
nginx-deployment	2/2	2	2	10s

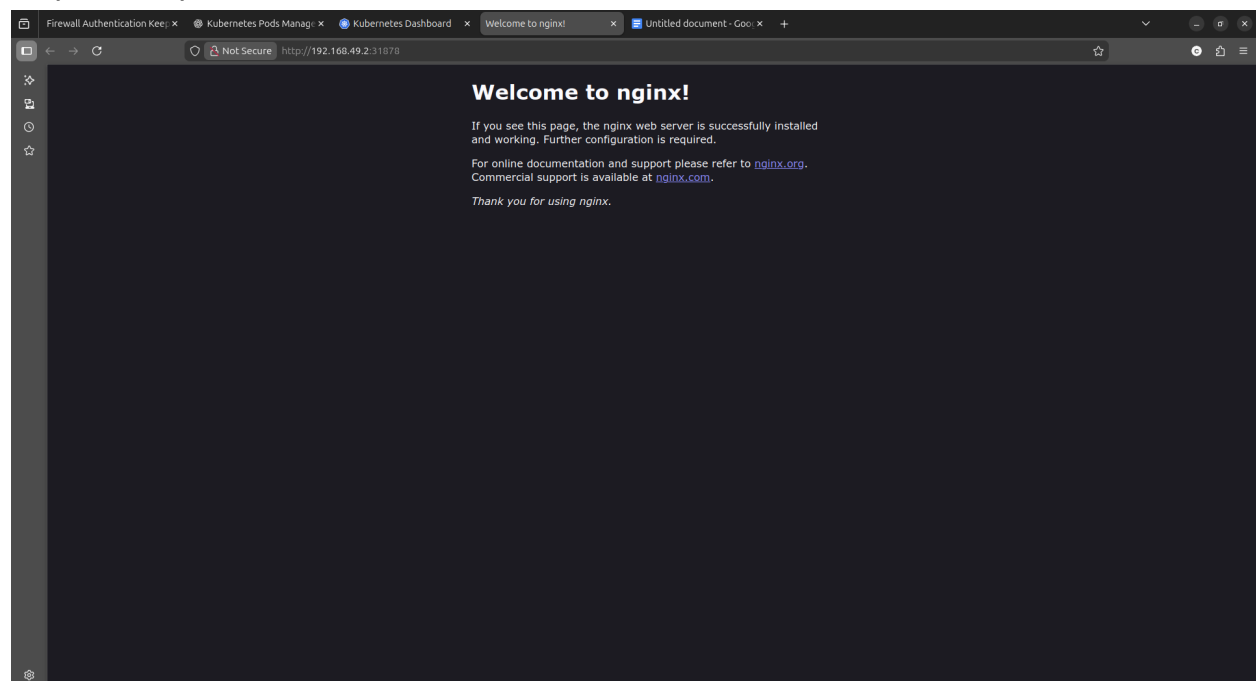
Step 5:- Start nginx service

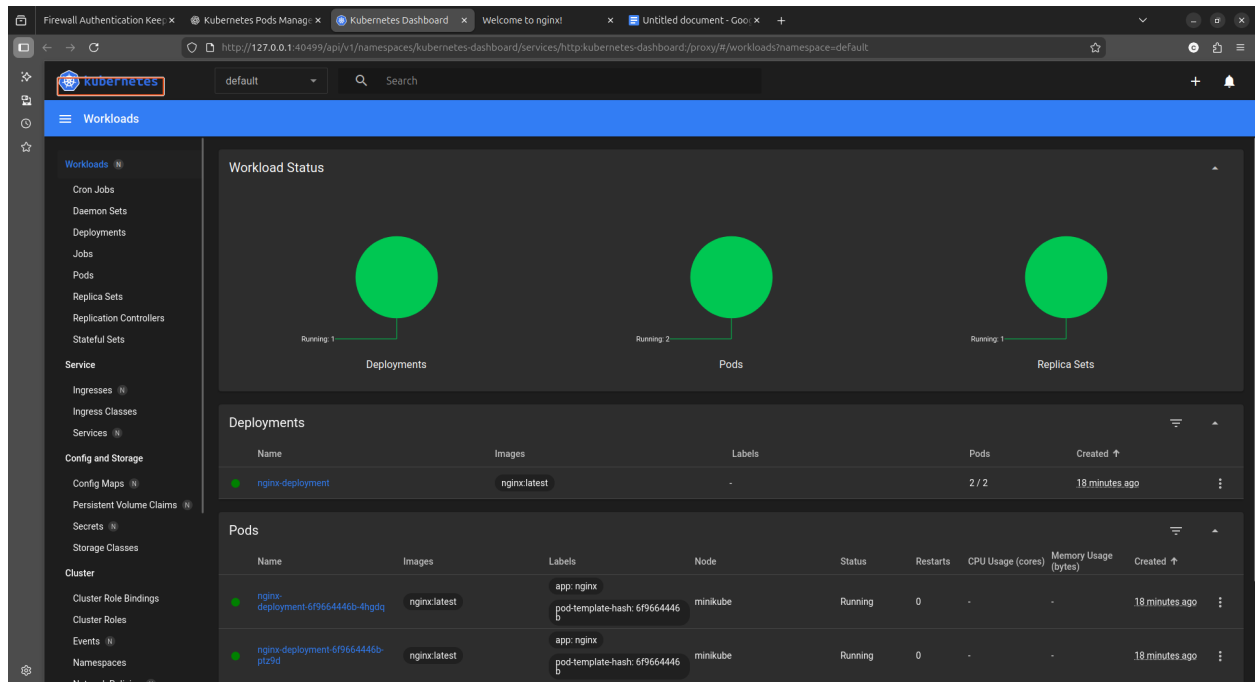
```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog10$ minikube service nginx-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:31878

🌈 Opening service default/nginx-service in default browser...

Step 6:- Output





Step 7:- Cleanup

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog10$ kubectl delete -f deployment.yaml
kubectl delete -f service.yaml
deployment.apps "nginx-deployment" deleted from default namespace
service "nginx-service" deleted from default namespace
```